Exploring the Limits of Machine-Learning Classification of Neutron-Star Matter Models:

Populations of Equations of State, Literature-Anchored Errors and Calibrated Uncertainty

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Abstract

Machine-learning classification of neutron-star composition is usually demonstrated on synthetic data in which every composition is a single equation of state (EoS) and test points are random draws along the same sequences. The initial audit shows that the headline 97% accuracy of a released benchmark of 770 such models is largely interpolation between neighbouring models (the MLP accuracy falls from 97% to 94% and 69% when blocks of six and 24 models are withheld) and that its tabulated surface redshift leaks class information. The benchmark is replaced by a population study of 360 EoS, 90 in each of four composition scenarios (nucleonic, hyperonic-like, strange, and neutron-decay dark-matter-admixed), solved with a full general-relativistic TOV, tidal and f-mode code that agrees with six cross-code reference f-mode frequencies (from an unpublished independent-code comparison) to 0.007%. With whole-EoS hold-out, six classifiers (logistic regression, k-NN, random forest, gradient boosting, SVM and an MLP ensemble) reach only 64% accuracy even without measurement error, and a finite-reference kernel classifier performs similarly: within these populations the plateau appears to be set by the information in the observables, not by the learner. Strange matter is separable (100% recall when f is measured accurately), hyperonic-like matter partly (74%), but nucleonic and dark-matter-admixed matter are indistinguishable once the nuclear EoS is unknown (pairwise accuracy 52%); with the nuclear EoS known exactly the dark component, which shifts f by about 0.56%, would need a $\sim$0.51% f-mode measurement to be detected at 75%. With errors anchored on published measurements and forecasts (NICER-like 10% in M and 9% in R, and a 20% f-mode measurement) the accuracy is 34–37% (chance 25%) ; the EoS-group class-conditional method has 94.8% empirical coverage with an average of 3.49 classes per set; 5%, 5% and 2% errors would give 53%, and the damping time adds no measurable classification gain in these tests. Probability calibration errors remain small in these synthetic-error tests, but the physical class ambiguity is substantial. Repeating the analysis on an independently constructed population (different parametrisations and priors) reproduces the nucleonic–dark-matter degeneracy (51%) and the NICER-like accuracy (37%), whereas the 64% plateau and the identifiability of the hyperonic-like family depend on how that family is built. This is a controlled forecasting and identifiability study, not an inference about observed stars. All conclusions remain conditional on the phenomenological EoS families, population priors, perturbative response and idealised error models.

1 Introduction

Neutron stars probe matter at densities well above nuclear saturation (Lattimer & Prakash, 2004; Özel & Freire, 2016), and a wide range of compositions—purely nucleonic matter, hyperon-rich matter, deconfined quark matter and neutron decay into a dark sector (Husain et al., 2022)—has been proposed. Machine learning has been applied to neutron-star inference, for example to map observed masses and radii to the EoS (Fujimoto et al., 2018). A natural question is whether a supervised classifier learns matter composition or the restricted EoS parametrisations encountered during training.

Previous full-GR studies of neutron-decay dark-matter stars have already demonstrated a degeneracy between hadronic and dark-matter-admixed f-mode observables (Shirke et al., 2024). Bayesian neural networks have likewise been used to infer dense-matter properties from neutron-star

Table 1: Composition of the released single-sequence data set (770 models). $N_{\geq 0.5}$ and $N_{\geq 1}$ are the numbers of models with $M \geq 0.5\,M_\odot$ and $M \geq 1\,M_\odot$.

Class	N	M range ($M_\odot$)	R range (km)	$M < 0.5\,M_\odot$	$N_{\geq 0.5}$	$N_{\geq 1}$
Nucleonic	222	7.2×10^{-8}–2.16	0.06–11.11	47%	118	83
Strange	220	4.2×10^{-6}–1.94	0.11–11.06	45%	122	80
Dark-matter	122	1.7×10^{-3}–0.68	0.51–6.56	67%	40	0
Hyperonic	206	2.0×10^{-4}–1.74	0.31–10.56	44%	115	69

observables (Carvalho et al., 2023). The question addressed here is not whether dark matter can shift f-mode frequencies or whether ML can fit neutron-star properties; it is whether composition classifications survive EoS-population variation, whole-EoS holdout and observationally motivated uncertainty.

A previous version of this work (arXiv:2512.23758v3) trained a small multilayer perceptron (MLP) on seven observables of 770 synthetic stellar models, each class being one EoS, and reported a test accuracy of 97.4% on 116 held-out models. That number does not support the conclusions drawn from it, and a single sequence per class cannot say how well compositions can be told apart when the EoS itself is uncertain. This revision therefore does the following.

1. **Audits the single-sequence benchmark** (Sec. 2): class sizes, mass coverage, a class-dependent redshift inconsistency, and the dependence of the accuracy on how test points are chosen.

2. **Builds a population of EoS** for each composition scenario and solves every one with a validated full-GR code (Sec. 3); the validation against six independent f-mode frequencies is given in Table 4.

3. **Evaluates with whole-EoS hold-out**, six learners and a kernel-Bayes reference classifier (Secs. 4, 5), under error scenarios taken from published measurements and forecasts as well as a scale that shows what precision would be needed.

4. **Tests statistical and EoS-family robustness**: EoS-level bootstrap intervals, learning curves, empirical prediction-set coverage, dark-fermion stability-window restrictions and out-of-family transfer between two parametrisations.

Appendix A keeps the single-sequence noise study, and Appendix C lists each claim of the earlier version that has been withdrawn or changed.

2 What the single-sequence benchmark can and cannot show

2.1 Audit of the released data set

The released data (`DataML.xlsx`) hold 770 models of four classes: a nucleonic and a hyperonic quark–meson-coupling EoS (Rikovska Stone et al., 2007), a neutron-decay dark-matter scenario (Husain et al., 2022) and MIT-bag strange matter; the oscillation quantities come from the earlier oscillation code of Husain (2026a) and are used as released here. Table 1 and Fig. 1 summarise the data.

1. *Class sizes.* The classes contain 222, 220, 122 and 206 models (not ≈ 190 each).

2. *Sampling.* Models lie on a regular radius grid ($\Delta R = 0.05\,\mathrm{km}$) along each sequence, so neighbouring models are almost identical: for 96.5% of the models the nearest neighbour in standardised $(\ln M, \ln R, \ln f, \ln \tau)$ belongs to the same class.

3. *Mass coverage.* 49% of the models have $M < 0.5\,M_\odot$ and 70% have $M < 1\,M_\odot$, down to $7.2 \times 10^{-8}\,M_\odot$.

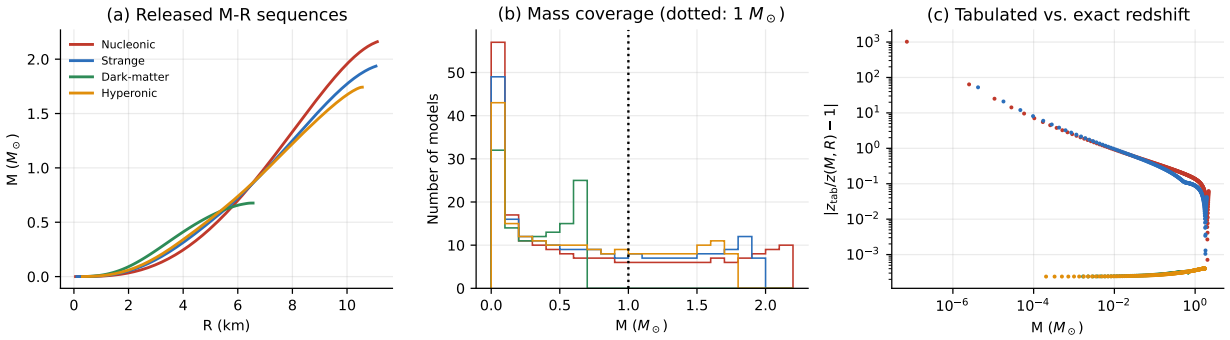


Figure 1: (a) Mass–radius sequences of the four classes. (b) Distribution of masses; the dotted line is $1\,M_\odot$. (c) Fractional difference between the tabulated surface redshift and the value implied by M and R; the hyperonic and dark-matter classes agree to $\sim$0.03%, the nucleonic and strange classes do not.

Table 2: Random-split test accuracy (%) over 100 seeds (stratified 70/15/15, $N_{\rm test} = 116$), noise-free.

Setup	Mean	s.d.	Min–max
Published setup: 7 columns incl. z, MLP	97.7	1.6	92.2–100.0
Published setup, random forest	99.2	1.1	94.0–100.0
Redshift removed, MLP	97.5	1.9	90.5–100.0
Redshift removed, random forest	98.9	1.2	94.0–100.0
Redshift removed, k-NN	98.7	1.0	96.6–100.0
M, R, f, τ only, MLP	93.6	3.5	79.3–99.1
M, R only, k-NN	60.1	4.1	50.9–69.8
M, R only, logistic regression	27.1	4.0	19.0–41.4
M only, logistic regression	25.3	3.4	15.5–31.9

4. *The dark-matter sequence ends at* $0.68\,M_\odot$, so at $M \gtrsim 0.7\,M_\odot$ the class is excluded by the mass alone.

5. *Redshift inconsistency.* For the dark-matter and hyperonic classes the tabulated z equals the Schwarzschild value implied by (M, R) to $\approx$0.031%; for the nucleonic and strange classes the median deviation is +18% and +11%. A logistic regression on the redshift residual alone separates {nucleonic, strange} from {dark matter, hyperonic} with 91% cross-validated accuracy (57% for M alone, 72% for M and R). The redshift is therefore a class-dependent artefact and is not used in any analysis here.

2.2 Replicating the earlier protocol

Table 2 and Fig. 2 repeat the earlier random-split protocol for 100 seeds. The published 97.4% is typical (97.7 ± 1.6%), so it is reproducible, but it is not specific to the MLP: a random forest reaches 99.2%. Removing the redshift changes the MLP result only from 97.7% to 97.5%. With M and R only, a k-nearest-neighbour classifier reaches 60.1%, and with M alone logistic regression is at chance (25.3%).

2.3 Leakage from neighbouring models

Figure 3 holds out contiguous blocks of increasing length (noise-free, 70/30 split, 15 repetitions). With all six observables (M, R, f, Q, τ, h_c) the MLP accuracy decreases from 97% (single-model holdout, equivalent to a random split) to 94% at six-model blocks and 69% at 24; the random forest drops from 99% to 82% and 42%. With the four plausible observables M, R, f, τ the MLP values are 78%, 69% and 45%. A large part of the apparent skill is therefore interpolation between near-duplicate neighbours of the same sequence. Q and h_c are model-derived amplitudes rather than measurable quantities and are not used further. Even with the leakage removed, a single

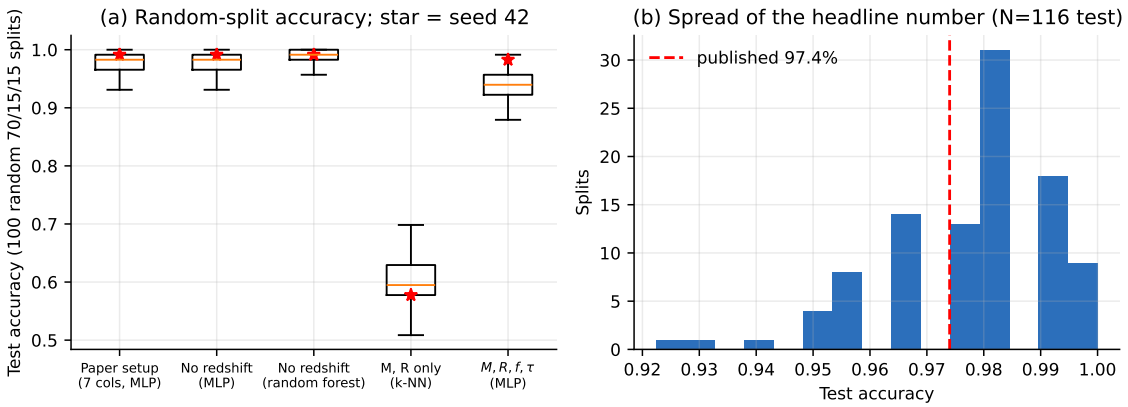


Figure 2: (a) Test accuracy over 100 random splits for several setups (star: seed 42, used in the earlier version). (b) Distribution for the original setup.

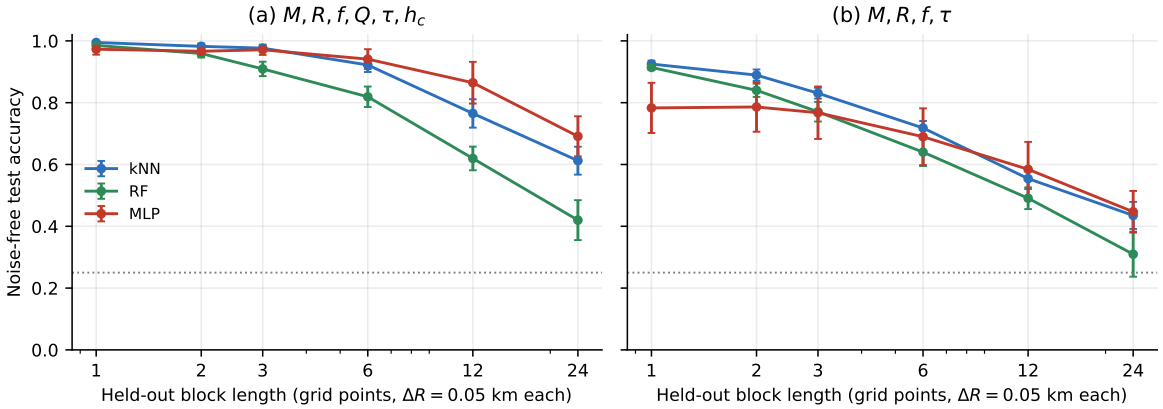


Figure 3: Noise-free test accuracy versus the length of the held-out contiguous mass block (mean and 95% t-interval over 15 repetitions). Dotted: chance. A block length of one is equivalent to the random split of the earlier version.

sequence per class cannot say whether the *composition* is identifiable when the EoS is uncertain: "classifying the family" and "recognising one EoS" are the same task. The remainder of the paper addresses this with populations.

3 A population of equations of state

3.1 Composition scenarios as EoS populations

Each scenario is a family of EoS defined by a small number of parameters with a uniform prior (Table 3); one draw from the prior is one EoS. The families are phenomenological parametrisations *inspired by* the four scenarios of the earlier version, not microphysical calculations. For the stable-dark-fermion benchmark, the relevant approximate mass interval is $937.9 < m_\chi < 938.783\,\mathrm{MeV}$ (Fornal & Grinstein, 2018). The exploratory population-A prior extends to $939.2\,\mathrm{MeV}$ and is therefore *not wholly physically admissible for that channel*. Out-of-window draws are treated solely as a sensitivity extension, not viable examples of the specified particle model. Section 5.7 reports the analysis restricted to the stability window and verifies which conclusions persist. Draws are accepted only if they satisfy broad observational and consistency screens: $M_{\max} \geq 1.97\,M_\odot$, a central sound speed satisfying $c_s^2 \leq 1$ at the maximum-mass configuration, $70 \leq \Lambda_{1.4} \leq 800$ and $10 \leq R_{1.4} \leq 14\,\mathrm{km}$. The acceptance rates are 21%, 57%, 24% and 26% for the nucleonic, strange, dark-matter and hyperonic-like priors, so the screens remove most of the prior volume (mainly through $M_{\max}$ and causality, and for strange matter through $\Lambda_{1.4}$ and $R_{1.4}$). Observables are

Table 3: Parametrisations and priors. PP: piecewise polytrope (Read et al., 2009) with core break densities $\rho_1 = 10^{14.7}$ and $\rho_2 = 10^{15}\,\mathrm{g\,cm^{-3}}$ on a four-piece crust; CSS: constant speed of sound, $\varepsilon = \varepsilon_0 + p/c_s^2$.

Scenario	Parameters (uniform priors)
Nucleonic	PP: $\log_{10} P_1 \in [34.15, 34.60]$ ($\mathrm{dyn\,cm^{-2}}$ at ρ_1), $\Gamma_1 \in [2.4, 3.8]$, $\Gamma_2 \in [2.4, 3.9]$, $\Gamma_3 \in [2.2, 3.9]$
Hyperonic-like	PP with $\log_{10} P_1 \in [34.20, 34.60]$, $\Gamma_1 \in [2.6, 3.8]$ and a soft segment $\Gamma_2 \in [1.8, 2.6]$ above $\rho_1 \approx 2n_0$ (hyperon onset), then a causal CSS segment with $c_s^2 \in [0.5, 1]$ above $\rho_t = 10^u$, $u \in [14.95, 15.25]$
Strange (self-bound)	CSS with $\varepsilon_0 = 4B$, $B^{1/4} \in [130, 160]\,\mathrm{MeV}$, $c_s^2 \in [1/3, 1]$ ($c_s^2 = 1/3$ is the MIT bag model)
Dark-matter admixed	nucleonic PP prior plus a self-interacting dark Fermi gas in chemical equilibrium with the baryons (neutron-decay scenario): $m_\chi \in [937.9, 939.2]\,\mathrm{MeV}$, repulsive self-interaction $G \in [50, 500]\,\mathrm{fm^2}$; effective one-fluid barotrope

tabulated for stars of 1.2–1.9 $M_\odot$ and evaluated on a mass grid over 1.2–1.85 $M_\odot$, where all accepted EoS have stable stars. The initial screening tested causality at the maximum-mass centre only; a later full-interior audit uncovered nine cases with superluminal sound speed at intermediate pressures. A separately rebalanced, fully causal sensitivity analysis is reported in Sec. 5.8.

The dark-matter-admixed scenario follows the effective-barotrope construction of the earlier work: baryons (nucleonic PP, $\mu_N = (\varepsilon + p)/n_B$) and a dark Fermi gas with $\mu_\chi = \sqrt{m_\chi^2 + k_F^2} + \hbar c G n_\chi$ share one chemical potential, and the total pressure and energy density are summed at that potential. Because the dark component appears only above $\mu_N = m_\chi$, the stellar surface is purely baryonic and the dark component is at most a few per cent of the energy density even at the highest densities. The dark fermion is assigned baryon number $B_\chi = 1$ in the baryon-conserving conversion benchmark. Writing $n_N = n_n + n_p$ for the nuclear baryon density, the conserved total baryon density is $n_{B,\mathrm{tot}} = n_N + n_\chi$; the quantity n_B in $\mu_N = (\varepsilon_N + p_N)/n_B$ denotes the *nuclear* density only. The conversion condition on the background is $\mu_N = \mu_\chi$; it does not itself determine the perturbative conversion timescale. By construction the DM scenario shares its nuclear prior with the nucleonic family. The classifier therefore tests whether the weak DM signature survives uncertainty in the baseline nuclear EoS.

3.2 Solvers and validation

TOV and tidal deformability. Stars are integrated with the Tolman–Oppenheimer–Volkoff equations (Tolman, 1939; Oppenheimer & Volkoff, 1939) and the tidal-deformability equation (Hinderer, 2008).

Full-GR f-mode. The $l = 2$ quasi-normal f-mode of the non-rotating, barotropic star is computed from the linearised Einstein equations in the Regge–Wheeler gauge. The interior equations were derived symbolically (not taken from the literature) from the linearised Einstein equations and the conservation law, reduced to four first-order equations, and solved with regular boundary conditions at the centre and a vanishing Lagrangian pressure perturbation at the surface; this is the same physical system as the Lindblom–Detweiler formulation (Detweiler & Lindblom, 1985). The exterior solution is matched to the Zerilli equation (Zerilli, 1970), and the complex eigenfrequency is the zero of the ingoing-wave amplitude; the real part gives f and the imaginary part τ. The exterior equations were checked against an independent symbolic derivation (residual 10^{-12}). The complex eigenfrequency is found by a secant search on the analytic ingoing amplitude, starting within 3% of the root.

Numerical error budget. Varying one numerical setting at a time (interior and exterior integration tolerances from 10^{-8} to 10^{-13}, the TOV tolerance, the surface cut-off, the starting

radius at the centre and the outer matching radius from 1500 to 8000 km) changes f by at most 3×10^{-8} in relative terms and τ by at most 5×10^{-7}, for SLy and for SLy with $G = 100\,\mathrm{fm}^2$ at $1.4\,M_\odot$. These are more than five orders of magnitude smaller than the median dark-matter shift of interest (median 0.56%). The remaining sensitivity is physical: changing the crust pressure scale by ±10% shifts f of the nuclear EoS by 0.20% but changes the dark-matter shift only from 1.313% to 1.338% (SLy, $G = 100\,\mathrm{fm}^2$), so the *shift* is robust to the crust treatment even though absolute frequencies are not.

Damping times. Direct independent solver benchmarks for the damping times of precisely these six stellar configurations were not available. The time dependence is $e^{i\omega t}$, so $\tau = 1/\mathrm{Im}\,\omega$ and a damped mode has $\mathrm{Im}\,\omega > 0$. The exterior machinery that produces $\mathrm{Im}\,\omega$ (Zerilli integration, projection onto asymptotic ingoing and outgoing solutions, complex root search) was benchmarked on a case with a published answer: with the horizon matched to purely ingoing waves it gives the fundamental $l = 2$ quasi-normal mode of a Schwarzschild black hole, $M\omega = 0.37366 + 0.08896\,i$, against $0.37367 + 0.08896\,i$ (Leaver, 1985), i.e. relative errors of 3e-05 (real part) and 8e-06 (imaginary part). The interior contribution to τ is not independently tested, so statements about τ (in particular that it adds no classification information) are conditional on the solver.

External consistency check for stellar damping times. The computed damping times were compared, without parameter refitting, against the nonrotating-star empirical relation of Benhar et al. (2004),

$$\tau_{\mathrm{BFG}} = \frac{R^4}{c M_{\mathrm{g}}^3}\left(0.087 - 0.271\frac{M_{\mathrm{g}}}{R}\right)^{-1}, \qquad M_{\mathrm{g}} = \frac{G_{\mathrm{N}} M_\star}{c^2}, \tag{1}$$

with R, M_{g} in km and c in $\mathrm{km\,s^{-1}}$. For the six nuclear EoS at $1.4\,M_\odot$ in Table 4, the solver results differ by 6.0–9.6% (median absolute deviation 8.7%) from this independently published scaling relation. This is a consistency check of the damping-time magnitude and compactness dependence, *not* an independent complex-eigenfrequency calculation for the same stars. The scientific interpretation of τ therefore remains conditional on the perturbation solver.

Adiabatic response of the dark-matter configurations. The perturbation equations use the equilibrium barotrope, i.e. chemical equilibrium is assumed to be maintained during the oscillation. As a bound on the opposite limit, f was recomputed with a frozen-composition adiabatic response (baryon and dark numbers separately conserved, both components compressed together) for 24 dark-matter EoS at $1.4\,M_\odot$: the median dark-matter shift is 0.64% with the equilibrium response and 0.63% with the frozen one, a difference of 1.1% of the shift (at most 0.021% of f). For the tested 24 configurations, the shift is insensitive at the reported level to the *common-displacement* frozen-versus-equilibrium prescription. This is not a bound for genuine two-fluid oscillations: independent displacements, entrainment, chemical-relaxation timescales and nuclear composition gradients are not included. Hence the subpercent DM signature remains conditional on the chosen perturbative model.

Separate-fluid hydrostatic check. As an additional background diagnostic, the nuclear and dark hydrostatic pressure gradients were integrated separately in a common spacetime metric for SLy, AP4 and MPA1 at $1.4\,M_\odot$ with $G = 100$ and $300\,\mathrm{fm}^2$. For these six configurations the maximum absolute radius difference from the effective one-fluid TOV construction was 0.211%; the total mass was matched to better than 3×10^{-5}%. This tests the equilibrium stellar *background* only, not the perturbative dynamics or the existence of distinct dark and nuclear normal modes. The numerical details and output are in `population/twofluid_background.py` and `results_twofluid_background.json`.

Table 4: Validation at $1.4\,M_\odot$ against six published piecewise-polytropic EoS fits. R and Λ were used to calibrate one crust parameter; f was held out.

EoS	R (km)	$R_{\rm ref}$	Λ	$\Lambda_{\rm ref}$	f (Hz)	$f_{\rm ref}$	$\Delta f/f$ (%)	τ (s)
SLy	11.7055	11.7061	295.5	295.6	1935.20	1935.06	+0.0072	0.1920
AP3	11.9712	11.9717	367.1	367.2	1848.51	1848.40	+0.0059	0.2094
AP4	11.3192	11.3197	247.5	247.6	2007.63	2007.50	+0.0062	0.1793
ENG	11.9695	11.9699	369.9	370.1	1845.65	1845.53	+0.0066	0.2100
MPA1	12.4339	12.4343	478.9	479.1	1745.68	1745.58	+0.0057	0.2341
WFF2	11.1245	11.1249	224.9	225.0	2047.47	2047.34	+0.0062	0.1729

Table 5: Population data set (90 EoS per scenario after screening): ranges of $M_{\rm max}$ ($M_\odot$), $R_{1.4}$ (km), $\Lambda_{1.4}$ and f at $1.4\,M_\odot$ (Hz).

Scenario	N	$M_{\rm max}$	$R_{1.4}$	$\Lambda_{1.4}$	$f_{1.4}$
Nucleonic	90	1.97–2.59	10.9–13.8	218–682	1614–2061
Strange	90	2.02–3.34	10.0–11.7	265–797	1558–1979
Dark-matter	90	1.97–2.66	10.8–13.7	199–685	1613–2098
Hyperonic	90	1.97–2.30	10.1–13.5	104–663	1625–2380

Validation. The crust is a single calibrated parameter (the absolute pressure scale of a four-piece SLy-like crust, 1.135 times the electron-gas value), fitted only to the radius and tidal deformability of six published EoS fits at $1.4\,M_\odot$ (Table 4); the radii are then reproduced to 0.0006 km and Λ to 0.05%. The f-mode frequencies were *not* used in any fit, and are reproduced to 0.007% (maximum over the six EoS). The reference values are those of the finite-EoS pilot of the earlier work (Husain 2026b), which used the separately developed RelModPy full-GR code (`https://github.com/fgittins/RelModPy`); the six values are listed in Table 4 so that the comparison can be repeated with any other code. The pilot manuscript is unpublished, so this is a comparison with an independent code rather than with published benchmark tables. In addition, the dark-matter frequency shift at fixed nuclear EoS (SLy, $G = 100\,\mathrm{fm}^2$, $1.4\,M_\odot$) is 25.7 Hz against 25.8 Hz in the reference, and a tabulated version of the same EoS agrees with the analytic one to 0.013% in f. Damping times of 0.17–0.23 s are obtained for the six EoS, as expected for a $1.4\,M_\odot$ f-mode; no independent reference values for τ were available.

3.3 The data set

Table 5 and Fig. 4 summarise the 360 EoS (2886 computed modes in total; about 202 s of computation per EoS). The strange family lies on a distinct f–R relation, whereas the nucleonic, dark-matter and hyperonic-like families overlap heavily; the hyperonic-like family extends to higher frequencies.

4 Methods

4.1 Whole-EoS validation

Every EoS belongs to exactly one of five folds (stratified by scenario). In each of five rounds one fold is the test set, the next fold is the calibration set and the other three are the training set; no EoS appears in more than one role, so every reported number is for EoS the classifier has never seen. The partition is repeated for three random seeds; the estimates are reported as means over folds within a repetition and Student-t 95% intervals over the three repetitions. Observations are (M, R, f, τ) at masses drawn uniformly over 1.2–1.85 $M_\odot$ (a mass prior shared by all classes), with independent noise draws, and classes are balanced (chance 25%).

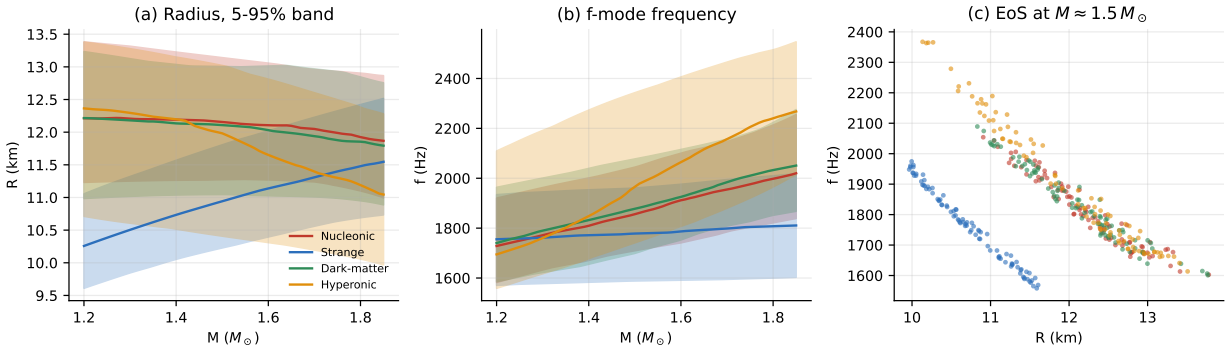


Figure 4: Population data set. (a) Radius and (b) f-mode frequency versus mass; lines are medians and bands the 5–95% range across EoS of each scenario. (c) f versus R at $1.5\,M_\odot$, one point per EoS.

4.2 Observational-error scenarios

Observables are perturbed multiplicatively, $x_j^{\rm obs} = x_j(1+\sigma_j\epsilon_j)$, $\epsilon_j \sim \mathcal{N}(0,1)$ independent. Draws with a non-positive factor (only possible for the damping time at large s: 0.6% of the draws at $s=2$ and 11% at $s=4$; none elsewhere) are redrawn, so the noise is positivity-truncated. Since the fractional standard deviations σ_j are fixed across the reference population, each Gaussian truncation factor $\Phi(1/\sigma_j)$ is independent of the reference EoS and cancels in class-posterior ratios; the kernel calculation therefore has the correct relative likelihood for positive observations. *Literature-anchored scenarios:* **NICER-like**: $\sigma_M = 10\%$, $\sigma_R = 9\%$, no f or τ; these are the 68% uncertainties of the NICER analyses of PSR J0030+0451 ($M = 1.34^{+0.15}_{-0.16}$ and $1.44^{+0.15}_{-0.14}\,M_\odot$, $R = 12.71^{+1.14}_{-1.19}$ and $13.02^{+1.24}_{-1.06}$ km; Riley et al. 2019; Miller et al. 2019). **NICER-like plus f-mode**: in addition $\sigma_f = 20\%$, motivated by the forecast that third-generation detectors could measure the f-mode frequency of a GW170817-like inspiral to within a few hundred Hz (Williams et al., 2022), i.e. roughly 10–25% of f. *Precision scale:* to show what would be required, the same observables are also evaluated with errors $s \times (5\%, 5\%, 2\%, 20\%)$ in (M, R, f, τ) for $s \in \{0.25, 0.5, 1, 2, 4\}$, and in the noise-free limit. These values are not forecasts for any instrument; they define a scale on which the literature-anchored scenarios can be placed. Three observable sets are used: $\{M, R\}$, $\{M, R, f\}$ and $\{M, R, f, \tau\}$.

4.3 Classifiers

Learners: logistic regression, k-nearest neighbours ($k = 25$), a random forest (200 trees, minimum leaf 5), histogram gradient boosting, an RBF support-vector machine (8000 training samples) and an MLP deep ensemble (Lakshminarayanan et al., 2017) of three 2×64 ReLU networks (the architecture of the earlier version) on log-transformed, standardised inputs. Training data contain three noise realisations of 33 masses for each training EoS.

4.4 Kernel-Bayes reference classifier

For a reference set of (EoS, M) points the posterior of class k for an observation $\mathbf{x}^{\rm obs}$ is

$$P(k \mid \mathbf{x}^{\rm obs}) \;\propto\; \frac{1}{N_k} \sum_{i \in k} \prod_j \mathcal{N}\Big(x_j^{\rm obs};\, x_{ij},\, (\sigma_j x_{ij})^2\Big)\,, \tag{2}$$

which is the Bayes classifier *for the finite discrete reference population under its nominal independent Gaussian error likelihood and prior*. At large noise, positive-only sampling changes the Gaussian normalisation by the factor $\Phi(1/\sigma_j)$, which is common to all reference EoS for each observable and cancels in the class posterior. The comparator is therefore Bayes-optimal *within the finite reference set* under the specified noise law, but not for an unknown continuous EoS population. The training EoS are used as the reference (the same information as the learners) and, separately, all

EoS except the test fold (a larger reference). Unlike the single-sequence oracle of Appendix A, this is not an upper bound, because the test EoS are not in the reference set; agreement across these methods suggests an information plateau for the sampled models but does not prove a universal upper bound.

4.5 Uncertainty quantification

Calibration: expected calibration error (ECE, 15 bins), negative log-likelihood (NLL) and Brier score, before and after temperature scaling fitted on the calibration EoS. *Epistemic uncertainty*: mutual information across ensemble members. *Empirically calibrated prediction sets*: prediction sets are formed using the split-conformal score $1 - \hat{p}(y \mid x)$ at target miscoverage $\alpha = 0.1$ (Vovk et al., 2005; Angelopoulos & Bates, 2023), calibrated on the calibration EoS. The exchangeable unit is the EoS: the scores of all stars of a calibration EoS (33 masses, two noise draws each) are pooled, the correction uses n equal to the number of calibration EoS (about 72), although scores from multiple stars per EoS are still pooled. This pooling does *not* yield exchangeable calibration scores at the EoS level, and replacing the sample count by an EoS count in the quantile alone does not establish the finite-sample split-conformal guarantee. The 90% target and observed coverage are therefore *empirical held-out-EoS diagnostics*, not distribution-free guarantees. The separately evaluated EoS-group construction below provides the class-conditional finite-sample interpretation under its stated assumptions. The observed coverage is marginal over stars from the sampled populations, not conditional on class or mass.

Class-conditional EoS-group conformal calibration. A second, explicitly group-aware analysis draws one stellar observation independently from each calibration EoS according to the same mass/noise sampling law intended for a test star. For each calibration EoS e of class Y_e, the nonconformity score is $S_e = 1 - \widehat{p}_{Y_e}(X_e)$. For each class k, a separate threshold q_k is the $\lceil (n_k + 1)(1 - \alpha) \rceil$-th ordered value among the n_k calibration EoS scores of class k; if the rank exceeds n_k, the threshold is set to infinity. The prediction set is

$$\mathcal{C}(X) = \{k : 1 - \widehat{p}_k(X) \leq q_k\}. \tag{3}$$

Under exchangeability of independently generated EoS within each class, a future EoS drawn from the same conditional population, and one independently sampled mass/noise observation per EoS, this Mondrian split-conformal procedure has finite-sample *class-conditional marginal* coverage for a random test star. The guarantee is not simultaneous over every mass on the same EoS, conditional on an observed mass, or valid for a different underlying EoS population. Each five-fold split provides $n_k = 18$ calibration EoS in each class; for $\alpha = 0.1$, the discrete class-specific threshold is the largest of these 18 scores. Results are reported separately from the previously used pooled-score empirical diagnostic, which does not have this guarantee.

5 Results

5.1 Noise-free separability of EoS populations

Figure 5 shows the held-out-EoS accuracy without measurement error, and the left panel of Fig. 7 the confusion matrix. With M and R only the accuracy is 39% (MLP ensemble; logistic regression 36%, random forest 39%); adding the f-mode raises it to 64% and adding τ changes nothing (64%). All learners agree within a few points. The confusion matrix shows why the ceiling is so low: strange matter is recognised perfectly (100%), hyperonic-like matter in 74% of cases, but nucleonic and dark-matter-admixed stars are assigned to each other almost at random (recall 44% and 40%; accuracy between the two classes 52%, chance 50%). Merging them into one class gives 84% three-class accuracy; the baseline for this merged problem is 50% (always predicting the merged class, which holds half of the stars), not the 33% of three balanced classes. The reason

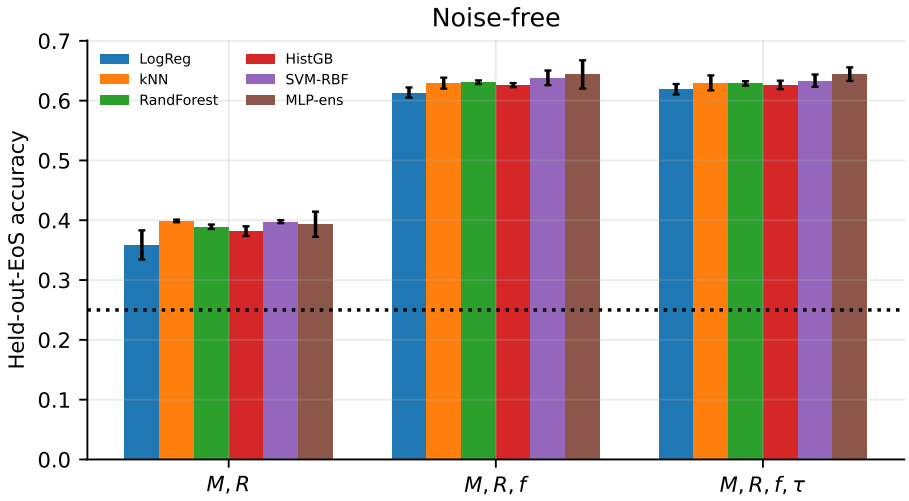


Figure 5: Held-out-EoS accuracy without measurement error for each learner and observable set (95% t-intervals over three repetitions; dotted: chance).

Table 6: Held-out-EoS accuracy (%, mean ± 95% t-interval over three repetitions) by error scenario and learner. Chance is 25%.

Scenario	LogReg	k-NN	RF	HistGB	SVM	MLP ens.	Kernel Bayes
NICER-like: M, R at 10%, 9%	33±1	32±1	31±1	33±1	33±1	34±1	34±1
+ f-mode at 20%	36±1	35±0	36±1	37±1	37±0	37±0	37±1
$s = 1$: 5%, 5%, 2%	51±1	52±1	53±1	53±2	53±2	53±1	53±2
$s = 1$ with τ 20%	51±1	52±0	52±1	53±1	53±2	53±1	53±2
$s = 0.25$ (1.3%, 1.3%, 0.5%, 5%)	59±0	60±1	60±1	60±2	60±2	61±2	60±2

is physical: the dark component shifts f by only 0.56% at fixed nuclear EoS (median over the population; 25.7 Hz or 1.3% for SLy with $G = 100\,\mathrm{fm}^2$; Sec. 5.5), far less than the spread of f at fixed mass produced by the uncertain nuclear EoS (about ±10%; Table 5).

5.2 Errors anchored on published measurements

Table 6 gives the accuracy under the literature-anchored scenarios and at selected points of the precision scale. With NICER-like errors (M 10%, R 9%) the accuracy is 34% (MLP ensemble; kernel-Bayes 34%), barely above the 25% chance level; adding a 20% f-mode measurement raises it to 37%. The three-class accuracy after merging nucleonic and dark-matter-admixed matter is 51%, no better than the 50% majority-class baseline of the merged problem. Errors of 5%, 5% and 2% in (M, R, f) would give 53%, and 1.3%, 1.3%, 0.5% would give 61%, which approaches but never exceeds the noise-free ceiling of 64%.

5.3 Precision scale and which observable matters

Figure 6 shows the accuracy against the precision scale. For the MLP ensemble, $\{M, R\}$ gives 39%, 37% and 29% at $s = 0.25$, 1 and 4, whereas $\{M, R, f\}$ gives 61%, 53% and 38%; the damping time with a 20% error (scaled with s) adds nothing within the uncertainty. Learners and kernel-Bayes agree to a few points at every scale (at $s = 1$: MLP 53%, kernel-Bayes 53%, kernel-Bayes with all other EoS as reference 53%; logistic regression 51%, random forest 52%, gradient boosting 53%, SVM 53%), so none of the learners tried leaves a measurable amount of information unused, and enlarging the reference sample from the training EoS to all other EoS does not change the result (this is evidence for an information plateau within these priors, not a proof of an upper bound).

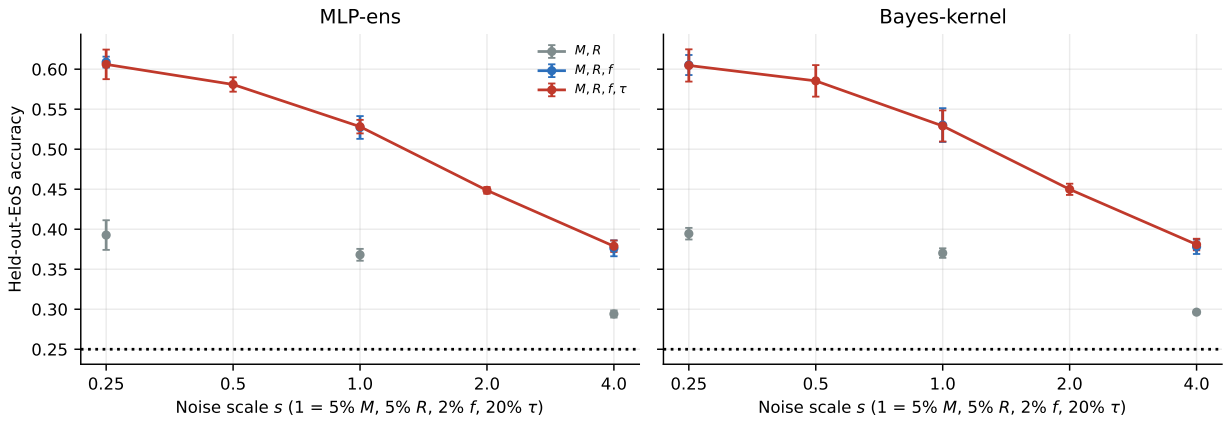


Figure 6: Held-out-EoS accuracy versus precision scale s ($1 = 5\%, 5\%, 2\%, 20\%$ in M, R, f, τ) for the MLP ensemble and the kernel-Bayes classifier, for three observable sets (95% t-intervals; dotted: chance). The $\{M, R, f\}$ and $\{M, R, f, \tau\}$ curves coincide.

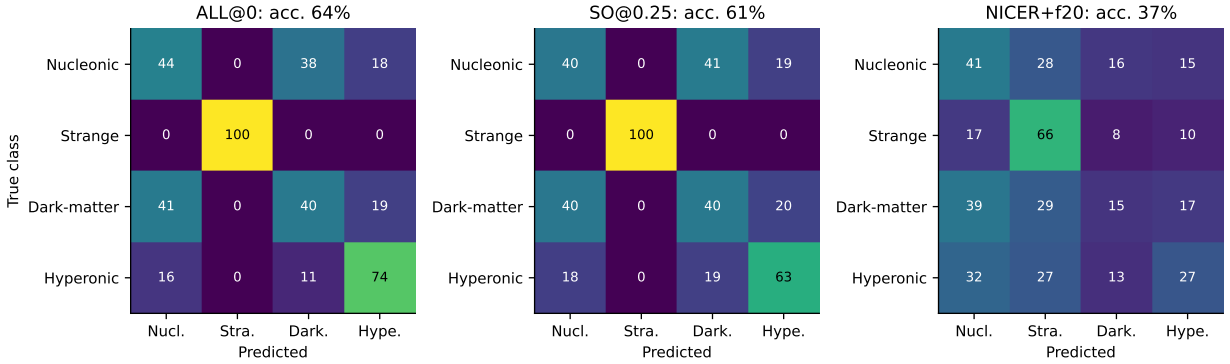


Figure 7: Row-normalised confusion matrices (%) of the temperature-scaled MLP ensemble, pooled over folds and repetitions. Left to right: noise-free, $s = 0.25$, and NICER-like errors plus a 20% f-mode measurement.

5.4 Degeneracies

Figure 7 shows row-normalised confusion matrices. With accurate f ($s = 0.25$) strange matter is still identified perfectly (100%) and the hyperonic-like class in 63%, but the nucleonic and dark-matter rows remain mixed (40% and 40% correct, 41% and 40% assigned to each other). With a 20% f-mode error all rows are confused with all columns, strange matter being the best recovered (66%).

5.5 Dark matter when the nuclear EoS is known

To separate "the dark component is small" from "the nuclear EoS is unknown", the observables of each of the 90 dark-matter-admixed EoS were also computed for its own nuclear base EoS without the dark component, at the same masses. At fixed nuclear EoS and mass the dark component raises f by a median of 0.56% (5–95% range 0.27–2.18%; positive for 100% of the pairs) and lowers R by a median of -0.40% (-1.46 to -0.20%); the shift falls steeply with the self-interaction (Fig. 8a; correlation of $\ln G$ with $\Delta f/f$ at $1.4\,M_\odot$ is -0.95). If the observer were told the nuclear EoS exactly, the Bayes accuracy for dark matter versus no dark matter (equal priors, independent Gaussian errors, mass identical by construction) would reach 75% at $\sigma_f = 0.51\%$ when R is known to 1% (0.45% for 5%) and 90% at $\sigma_f = 0.20\%$ (Fig. 8b); at the points of the precision scale it is 75% at $s = 0.25$ (0.5% in f, 1.25% in R), 58% at $s = 1$ and 52% at $s = 4$, and with a 20% f-mode error and 9% in R it is 51%. With the nuclear EoS *unknown* the accuracy between the nucleonic

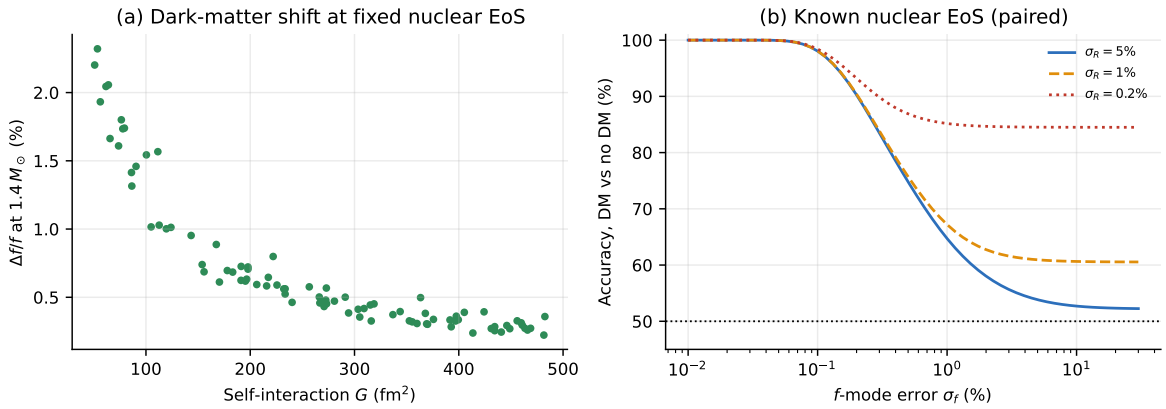


Figure 8: Dark matter at fixed nuclear EoS (90 paired EoS). (a) Fractional f-mode shift at $1.4\,M_\odot$ versus the self-interaction G. (b) Bayes accuracy for dark matter versus no dark matter, averaged over EoS and masses, as a function of the f-mode error for three radius errors (dotted: chance).

Table 7: Probability calibration and *pooled-score* empirical prediction sets (mean over three repetitions): ECE and NLL (nats) for the raw MLP ensemble, temperature scaling ($+T$) and finite-reference kernel, and pooled-score coverage and set size. This pooled-score procedure does not have a formal EoS-group split-conformal coverage guarantee; the class-conditional grouped alternative is reported in Table 8.

	ECE			NLL			MLP $+T$		Kernel Bayes	
Scenario	Raw	$+T$	KB	Raw	$+T$	KB	Cov.	Size	Cov.	Size
NICER-like	0.021	0.023	0.019	1.32	1.32	1.32	0.91	3.29	0.91	3.29
$+$ f at 20%	0.021	0.022	0.020	1.29	1.29	1.29	0.92	3.22	0.92	3.21
$s = 0.25$	0.053	0.043	0.093	0.71	0.71	0.83	0.91	1.85	0.92	1.93
$s = 0.5$	0.043	0.039	0.054	0.75	0.75	0.77	0.91	1.93	0.91	1.97
$s = 1$	0.038	0.037	0.039	0.92	0.92	0.92	0.91	2.24	0.91	2.24
$s = 2$	0.031	0.033	0.026	1.15	1.15	1.15	0.92	2.84	0.92	2.83
$s = 4$	0.026	0.028	0.021	1.29	1.29	1.29	0.91	3.24	0.91	3.23

and dark-matter classes is 52% even without any measurement error (Sec. 5). The degeneracy therefore comes from the uncertainty of the nuclear EoS, not from the smallness of the shift alone: detecting the dark component needs both an f-mode measurement at the 0.5% level and a nuclear EoS known to better than the shift itself.

5.6 Calibration and prediction sets

The reported confidence estimates have low empirical calibration error in these tests (Fig. 9a): at $s = 1$ the ECE is 0.038 for the raw ensemble, 0.037 after temperature scaling and 0.039 for kernel-Bayes, with NLL 0.92 nats for all three (Table 7). The empirically calibrated prediction sets have observed coverage between 0.91 and 0.92 at the 0.90 target for every scenario, under the EoS-count quantile correction (without a formal group-conformal guarantee). Their mean set size is 3.29 of four classes under NICER-like errors, 3.22 when a 20% f-mode is added, 2.24 at $s = 1$ and 1.85 at $s = 0.25$. Temperature scaling and kernel-Bayes give the same sets. At realistic precision the data therefore support a short list containing most of the classes, not a single label.

EoS-group conformal coverage. For each of three independently sampled fold partitions, the procedure of Sec. 4.5 is evaluated on all five held-out folds with one independently selected calibration observation per EoS (18 calibration EoS per class). The NICER-like scenario with a 20% f-mode error has mean marginal coverage 94.8% with an average of 3.49 candidate classes per set; the $s = 1$ precision scenario gives 94.4% and 2.55 classes (Table 8). The class-specific mean coverages are also shown. These wider sets make explicit the substantial ambiguity in the

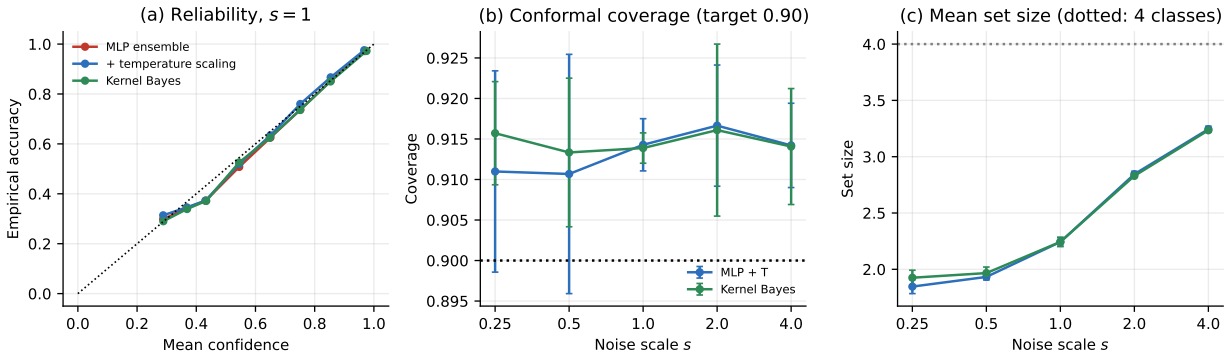


Figure 9: (a) Reliability diagram ($s = 1$, pooled). (b) Pooled-score empirical coverage and (c) mean prediction-set size versus precision scale (95% t-intervals).

Table 8: EoS-group Mondrian prediction sets at target miscoverage $\alpha = 0.1$, using 18 calibration EoS per class. All entries are means over five held-out-EoS folds and three independent partitions. N = nucleonic, S = self-bound strange, DM = dark-matter admixed, H = hyperonic-like. Coverage values are percentages; size is the mean number of admitted matter labels (out of four).

Scenario	Cov. (%)	Size	N (%)	S (%)	DM (%)	H (%)
NICER-like $+f$ (20%)	94.8	3.49	94.9	95.3	93.8	95.2
$s = 1$ precision	94.4	2.55	93.5	95.9	94.1	94.2

model classification. The difference from the more compact pooled-score sets in Table 7 illustrates why grouped calibration matters; these results remain conditional on exchangeability within the specified phenomenological EoS classes.

5.7 Statistical robustness

EoS-level bootstrap. The intervals of the preceding sections describe re-partitioning of the same 360 EoS. Table 9 instead gives 95% intervals from resampling whole EoS within each class (2000 resamples; each EoS's predictions pooled over the three partitions). The noise-free accuracy of the MLP ensemble is 64% [61, 67], the recall of strange matter is 100% [100, 100], and the accuracy between the nucleonic and dark-matter classes is 51% [45, 56]; with NICER-like errors and a 20% f-mode it is 50% [48, 53] and the overall accuracy 37% [36, 38]. The intervals for the nucleonic–dark-matter pair contain 50% in every scenario, so indistinguishability is not an artefact of a particular partition of this finite sample.

Population size. Figure 10 shows the accuracy as a function of the number of training EoS (54 to 216). Without noise the logistic regression is already at 61% with 54 training EoS and the MLP ensemble at 62% with 108, against 64% with 216; under NICER-like errors plus a 20% f-mode the accuracy is 35–37% at every size. The plateau therefore does not appear to result from too few EoS, and a larger population of the same kind is unlikely to change the numbers.

Dark-fermion mass window. Restricting the main population to the dark-matter EoS with $m_\chi \leq 938.783\,\mathrm{MeV}$ (63 per class, the stability window; Table 3) leaves the results unchanged: 64% without noise, 60% at $s = 0.25$, 52% at $s = 1$ and 37% for NICER-like errors plus a 20% f-mode, with an accuracy of 51% between the nucleonic and dark-matter classes.

5.8 Nuclear microphysics benchmarks and a full-interior causality audit

Published nuclear EoS anchor points. A separate sensitivity calculation uses six published piecewise-polytropic representations of nuclear microphysical models (SLy, AP3, AP4, ENG, MPA1

Table 9: EoS-level bootstrap estimates (%, 95% interval): overall accuracy, accuracy between the nucleonic and dark-matter classes (chance 50%) and recall of strange matter.

Scenario	Learner	Accuracy	Nucleonic vs DM	Strange recall
ALL 0	MLP-ens	64 [61, 67]	51 [45, 56]	100 [100, 100]
ALL 0	LogReg	62 [59, 65]	50 [44, 56]	100 [100, 100]
ALL 0	RandForest	63 [60, 65]	53 [49, 58]	100 [100, 100]
NICER+f20	MLP-ens	37 [36, 38]	50 [48, 53]	67 [65, 68]
NICER+f20	LogReg	36 [34, 37]	50 [47, 54]	61 [59, 62]
NICER+f20	RandForest	35 [35, 36]	50 [49, 51]	58 [57, 60]
NICER+f20	Bayes-kernel	37 [36, 39]	51 [48, 53]	67 [65, 68]
SO 1.0	MLP-ens	53 [51, 55]	49 [46, 52]	91 [89, 92]
SO 1.0	LogReg	51 [48, 53]	49 [45, 54]	90 [89, 91]
SO 1.0	RandForest	53 [51, 54]	50 [48, 51]	89 [88, 90]
SO 1.0	Bayes-kernel	53 [51, 55]	49 [46, 52]	91 [89, 92]
SO 0.25	MLP-ens	60 [58, 63]	49 [45, 54]	100 [100, 100]
SO 0.25	LogReg	58 [56, 61]	49 [44, 55]	100 [100, 100]
SO 0.25	RandForest	60 [58, 62]	50 [47, 53]	100 [100, 100]
SO 0.25	Bayes-kernel	61 [58, 63]	51 [48, 55]	100 [100, 100]

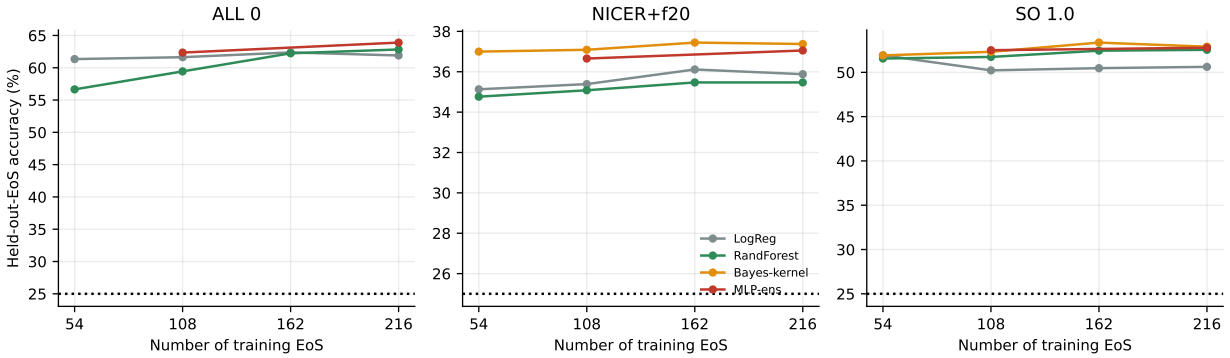


Figure 10: Held-out-EoS accuracy versus the number of training EoS for four learners (pooled over repetitions; dotted: chance). The MLP ensemble is shown at 108 and 216 training EoS only.

and WFF2) as baseline EoSs. For each, a $m_\chi = 938\,\mathrm{MeV}$ dark component is introduced using the same conversion prescription, with $G = 100$ and $300\,\mathrm{fm}^2$. At $1.4\,M_\odot$ the nuclear baseline frequencies span 1746–2047 Hz; the median DM-induced frequency changes are 1.215% (range 1.144–1.326%) for $G = 100\,\mathrm{fm}^2$ and 0.400% (range 0.380–0.442%) for $G = 300\,\mathrm{fm}^2$. Every sampled admixed configuration supports a maximum mass above $1.97\,M_\odot$. These externally motivated *nuclear baseline* checks support the scale and self-repulsion dependence of the modelled frequency changes, but they are not a population-level classifier evaluation on independent microscopic hyperon or quark EoSs, nor an independent eigenfrequency-solver validation.

Full-interior causality audit. The original selection checked c_s^2 at the maximum-mass centre but not every interior pressure. Re-evaluating c_s^2 along the full maximum-mass configuration identifies nine noncausal EoSs: five nucleonic and four DM-admixed models; none of the 90 strange or 90 hyperonic-like models fail the adopted tolerance 10^{-3}. A class-balanced resampling of 85 EoSs per family (340 total), followed by whole-EoS validation, yields MLP mean accuracy 63.1% without measurement error, compared with 64% in the original sample; under NICER-like errors plus a 20% f-mode measurement, accuracy is 36.7% versus approximately 37%. The conditional nucleonic-versus-DM accuracy is 49.0% in the noise-free rebalanced sample. These tests show that the principal reported degeneracy is not driven by the nine noncausal members. However, the original main-sample metrics retain those members for auditability, and a future reference population should impose causality at *all* relevant densities before sampling. The checks are

Table 10: Alternative population B. ε_x is the energy density at $1.3\,n_{\rm sat}$ ($\rho_x = 10^{14.55}\,\mathrm{g\,cm^{-3}}$); sound-speed profiles are piecewise linear in $\ln\varepsilon$ through knots at $\varepsilon/\varepsilon_x = 1, 1.5, 2.5, 4, 7$ and constant above; the lower-density EoS is a single polytrope on the same crust.

Scenario	Parameters (uniform priors)
Nucleonic	polytrope $\log_{10} P_1 \in [34.15, 34.55]$, $\Gamma_1 \in [2.6, 3.6]$ up to ρ_x; then $c_s^2 \in [0.1, 1]$ at each knot (first knot $[0.1, 0.6]$)
Hyperonic-like	as nucleonic, but $c_s^2 \in [0.15, 0.6]$ at the first two knots, a dip $c_s^2 \in [0.03, 0.3]$ at $2.5\,\varepsilon_x$ and $c_s^2 \in [0.5, 1]$ at the last two
Strange (self-bound)	$P(\mu) = 3a_4\mu^4/(4\pi^2) - 3a_2\mu^2/(4\pi^2) - B$ with $a_4 \in [0.55, 1]$, $\sqrt{a_2} \in [0, 150]\,\mathrm{MeV}$, $B^{1/4} \in [135, 165]\,\mathrm{MeV}$
Dark-matter admixed	nucleonic B plus a dark Fermi gas with $m_\chi \in [937.9, 938.78]\,\mathrm{MeV}$ (inside the stability window) and $G \in [50, 500]\,\mathrm{fm}^2$

Table 11: Population B (48 EoS per scenario): ranges of $M_{\rm max}$ ($M_\odot$), $R_{1.4}$ (km), $\Lambda_{1.4}$ and f at $1.4\,M_\odot$ (Hz).

Scenario	N	$M_{\rm max}$	$R_{1.4}$	$\Lambda_{1.4}$	$f_{1.4}$
Nucleonic	48	2.04–3.05	11.5–13.5	309–795	1563–1913
Strange	48	1.97–2.20	10.9–11.7	441–761	1575–1776
Dark-matter	48	2.01–3.05	11.6–13.7	287–798	1557–1947
Hyperonic	48	2.02–2.51	11.9–13.5	290–797	1556–1942

reproducible from `audit_causality.py` and `causal_filtered_classification.py`.

5.9 Robustness to the EoS populations

The preceding results depend on the parametrisations and priors of Table 3. To test this, an independent population ("B") was constructed with different functional forms and priors (Table 10); the same screens, solver and experiments were applied (48 EoS per scenario, 192 in total; Table 11). Population B is still parametric rather than microphysical, so it tests sensitivity to the parametrisation, not to the underlying nuclear physics.

Within population B. With whole-EoS hold-out, the MLP ensemble reaches 49% without noise (logistic regression 52%, random forest 52%), against 64% for population A. Strange matter is again recognised perfectly (100%) and the accuracy between the nucleonic and dark-matter classes is again at the level of chance (51%); nucleonic matter is recognised in 34%, dark-matter in 37% and hyperonic-like matter in only 26% (74% in population A), so the separability of the hyperonic-like family, and with it the overall plateau, depends on how the family is built. At $s = 0.25$ and $s = 1$ the accuracies are 50% and 45% (A: 61% and 53%), and for NICER-like errors plus a 20% f-mode 37% (A: 37%); there strange matter is recognised in 74% of cases.

Out-of-family transfer. Table 12 gives the accuracy of classifiers trained on all EoS of one population and tested on all EoS of the other. Without noise the transfer accuracy is 52% (A→B) and 54% (B→A) for the MLP ensemble, between the within-population values, and with NICER-like errors plus a 20% f-mode 36% and 33%. Strange matter is still recognised in 100% and 100% of cases without noise, and the accuracy between the nucleonic and dark-matter classes remains 50% and 56%. The results that survive a change of population are therefore (i) the nucleonic–dark-matter degeneracy, (ii) an accuracy of about 33–37% under NICER-like errors, with or without a 20% f-mode, and (iii) the identifiability of self-bound strange matter once f is accurate; the 64% plateau and the identifiability of the hyperonic-like family are specific to the construction of population A.

Table 12: Out-of-family test: accuracy (%, mean ± 95% t-interval over three noise realisations) of classifiers trained on all EoS of one population and tested on all EoS of the other. Chance is 25%.

Train→test	Scenario	LogReg	RF	HistGB	MLP ens.	Kernel Bayes
A→B	ALL 0	52±0	50±0	50±7	52±5	–
A→B	NICER+f20	35±1	34±1	36±1	36±1	36±1
A→B	SO 1.0	42±0	38±2	39±1	39±1	39±0
A→B	SO 0.25	52±0	47±1	46±2	52±1	46±0
B→A	ALL 0	51±0	50±2	51±0	54±1	–
B→A	NICER+f20	31±1	34±1	34±1	33±1	34±1
B→A	SO 1.0	40±0	46±1	46±1	48±1	46±1
B→A	SO 0.25	50±0	51±2	51±2	54±1	51±0

6 Discussion and limitations

What the results support. (i) A high accuracy on single-sequence data does not carry over to populations: with whole-EoS hold-out the highest accuracy reached by any method for these four scenarios is about 64% even without measurement error (49% in the alternative population B). (ii) The plateau looks informational: six different learners (including the MLP ensemble) and a kernel-Bayes classifier built on the exact error model agree, so within these priors a better learner is unlikely to help (the kernel-Bayes classifier uses a finite reference population and is not a rigorous upper bound). (iii) With the errors of current X-ray measurements and a plausible third-generation f-mode measurement, composition is essentially unidentifiable (34–37% against 25% chance); an accurate f-mode is the observable that would help, and the damping time adds no measurable classification improvement in the tested scenarios. (iv) Dark-matter-admixed matter is degenerate with nucleonic matter unless the nuclear EoS is known to better than the shift in f that the dark component produces (median 0.56%, up to about 2% for weak self-interaction); with the nuclear EoS known exactly, an f-mode precision of about 0.51% would give 75% accuracy (Sec. 5.5), in line with the EoS-conditional analysis of the earlier pilot. (v) Probability calibration and class-conditional EoS-group conformal prediction quantify the residual ambiguity: grouped coverage is 94.8% in the NICER-like plus 20% f-mode scenario, at the cost of admitting 3.49 classes on average. The original pooled-score sets lack the group-conformal guarantee.

Limitations. *Phenomenological families.* The EoS families are parametrisations with uniform priors, screened by $M_{\max}$, causality, $\Lambda_{1.4}$ and $R_{1.4}$; conclusions depend on these priors. The nucleonic–dark-matter degeneracy in particular follows from the dark-matter scenario sharing the nuclear prior. Microphysical models (for example a population of QMC EoS) would change the priors and might make some classes easier or harder to separate. The hyperonic-like family is a soft-then-stiff parametrisation, not a hyperonic calculation, and the strange family uses a constant speed of sound; the classifier therefore separates parametrisation families. A second population with different functional forms and priors (Sec. 5.9) preserves the nucleonic–dark-matter degeneracy, the NICER-like accuracy and the recognition of strange matter (self-bound models with a surface density jump give a distinct f–R relation in both), but not the 64% plateau or the identifiability of the hyperonic-like family. Six published polytropic representations of nuclear microphysical EoSs now provide additional nuclear baseline checks (Sec. 5.8); these do not replace a population of self-consistent microphysical hyperonic, quark and dark-matter stellar models. The principal composition-classification results remain conditional on the constructed phenomenological families. *Dark matter.* The dark component is treated as an effective one-fluid barotrope in chemical equilibrium; separate-fluid hydrostatic integration agrees on stellar backgrounds to within 0.211% in six checked configurations, but is not a two-fluid eigenmode calculation; the check against the two-fluid pilot is a single number (25.7 versus 25.8 Hz), and a frozen-composition response changes the shift by only 1.1% (Sec. 3.2). Independent displacement of the two fluids (a genuine two-fluid eigenmode calculation), other dark-sector models and other dark fractions are not explored. *Solver.*

Non-rotating $l = 2$ stars are considered. The main calculation assumes the equilibrium-barotrope response; the 24-EoS test varies that response under a shared displacement, not full two-fluid motion. Composition gradients, entrainment and crust elasticity are omitted. Black-hole QNM benchmarks test the exterior complex-frequency machinery, but independently published stellar damping-time benchmarks are still unavailable. *Errors.* Independent multiplicative noise is used, with resampling of nonpositive factors at high errors; its class-independent truncation normalisation cancels from finite-reference kernel posterior odds. NICER and f-mode uncertainties are motivated by individual-source measurements and forecasts, not a joint detected sample. Correlated errors, selection effects and systematic theoretical uncertainty are unmodelled. *Initial causality screen.* Nine EoSs from the original 360-model population violate the full interior sound-speed criterion despite passing a central-only check. The explicitly rebalanced 340-EoS test in Sec. 5.8 leaves the headline results essentially unchanged, but the main classification numbers were not recomputed for a freshly sampled, fully causal 360-EoS population.

Statistics and sampling. The class-conditional, one-score-per-EoS conformal procedure provides a finite-sample marginal guarantee for one randomly sampled future stellar observation when EoS are exchangeable within classes; it does not establish conditional coverage for specific masses or transfer to other priors. The earlier pooled-score sets are empirical only. The 90 EoS per scenario (48 in population B) and three fold partitions that reuse the same EoS, so the t-intervals reflect partition and noise variability only; EoS-level bootstrap intervals and learning curves are given in Sec. 5.7. A uniform mass prior over 1.2–1.85 $M_\odot$ is used. Fold-level class-specific coverage can vary substantially despite the good aggregate grouped coverage. *Observational realism.* The errors are independent and Gaussian; real NICER posteriors are correlated and non-Gaussian, and the GW170817 and NICER posteriors themselves were not used. *Not comparable to earlier data.* The new data come from a new solver and cannot be compared point by point with the single-sequence data set.

7 Conclusions

Evaluated on populations of EoS with whole-EoS hold-out, machine-learning classification of neutron-star composition is far less decisive than the 97% single-sequence benchmark suggested: the accuracy plateau is about 64% even without measurement error (49% in an alternative population with different parametrisations), it appears to be set by the information in the observables rather than by the learner, strange matter is the only scenario that is cleanly separable (in the family defined here), and nucleonic and dark-matter-admixed matter are not distinguishable when the nuclear EoS is unknown. With errors anchored on current X-ray measurements and a forecast f-mode precision the accuracy is 34–37%, and the class-conditional EoS-group prediction sets have 94.8% mean held-out coverage while admitting an average of 3.49 of four possible composition labels. Separately, published nuclear EoS anchor fits reproduce subpercent-to-percent dark-matter frequency shifts, and exclusion of nine originally accepted but noncausal EoSs leaves the reported classification degeneracy intact. These tests improve robustness within the chosen modelling framework, but independent microscopic composition populations and fully coupled two-fluid oscillation eigenproblems are still needed for model-general statements.

Data, code and reproducibility

The reproducible EoS-group calibration is implemented in `group_conformal.py` and `run_group_calibration.py`; the published damping-time scaling cross-check is in `tau_consistency.py`. The microphysics-derived nuclear baseline comparisons, separate-fluid hydrostatic checks, full-interior causality audit and causal-only classification are implemented in `microphysics_anchor.py`, `twofluid_background.py`, `audit_causality.py` and `causal_filtered_classification.py`. The single-sequence analysis uses `audit.py`, `repro.py`, `blocksize.py`, `main.py`, `extrap.py`, `interp_check.py` and `report.py`; the population study uses the solver and generator in `population/` (`eos.py`, `tov.py`, `fmode2.py`,

`families.py`, `gen_pop.py`, `gen_base.py`, `paired_analysis.py`, `families_b.py`, `stat_robust.py`, `cross_family.py`, `frozen_test.py`, `bh_qnm.py`, `robust_report.py`), the derivation scripts (`derive_vac2.py`, `derive_matter.py`, `elim.py`), the experiment (`pop_experiment.py`) and the report scripts. The source code, inputs and automated manuscript build are available on the `paper/referee-revision-2026-10-19` branch of the repository below; the new grouped-calibration outputs and damping-time comparisons are archived as reproducibility-workflow artifacts. A DOI-tagged immutable release remains to be deposited before journal submission; the convergence tests are in `convergence_tests.py`. The 360 generated EoS and all results are included as JSON/CSV; figures, tables and every number in the text are generated from them. Fixed seeds are used throughout. The environment is Python 3.13.16 with scikit-learn 1.9.1. The released single-sequence data are at `https://github.com/wasif553/ML-NeutronStar-Classification`.

A Single-sequence noise study (supplement)

This appendix keeps the noise, oracle and uncertainty analysis performed on the single-sequence benchmark, with blocked validation. It is superseded by the population study for questions about composition, but it isolates the effect of measurement error on the original data and provides an exact oracle bound.

Mass regimes, errors, learners, oracle

Three mass regimes are used: **A**, all models ($N = 770$); **B**, $M \geq 0.5\,M_\odot$ ($N = 395$, with 40 dark-matter models); **C**, $M \geq 1\,M_\odot$ ($N = 232$, three classes). Contiguous mass blocks of six models are assigned at random to training (60%), calibration (20%) and test (20%) roles; the analysis is repeated for 10 random assignments. Observables are perturbed as $x_j(1 + s\sigma_j\epsilon_j)$ with $(\sigma_M, \sigma_R, \sigma_f, \sigma_\tau) = (5, 5, 2, 20)\%$ at $s = 1$. The learner is a three-member MLP ensemble, with logistic regression and a random forest as baselines. The oracle Bayes classifier knows the exact sequences (including the test blocks) and the error model,

$$P(k \mid \mathbf{x}^{\rm obs}) \;\propto\; \frac{1}{N_k} \sum_{i \in k} \prod_j \mathcal{N}\Big(x_j^{\rm obs};\, x_{ij},\, (s\sigma_j x_{ij})^2\Big)\,, \tag{4}$$

and is an upper bound for learners trained on finite data.

Results

At $s = 1$ in regime B the ensemble reaches 47% with M, R, f, τ against an oracle ceiling of 65% (chance 25%); M and R alone give 29% (oracle 49%). In regime C (three classes) the values are 50% (ensemble) and 65% (oracle). Adding f raises the oracle from 49% to 64%; adding a 20% τ gives 65%. Strange and hyperonic matter are the most confused pair (36% of strange models labelled hyperonic and 37% the reverse). The raw ensemble is overconfident (ECE 0.188, NLL 1.13 at $s = 1$, regime B; oracle 0.040, 0.74); temperature scaling reduces them to 0.096 and 1.00. Conformal sets have about 90% marginal coverage and contain 2.35 of four classes at $s = 1$ (oracle 1.85). When the heaviest 25% of each class is withheld, the ensemble reaches 50% (oracle 80%) and the conformal sets cover only 0.81, since exchangeability fails; the ensemble's mutual information rises from 0.010 to 0.025 nats, an incomplete warning.

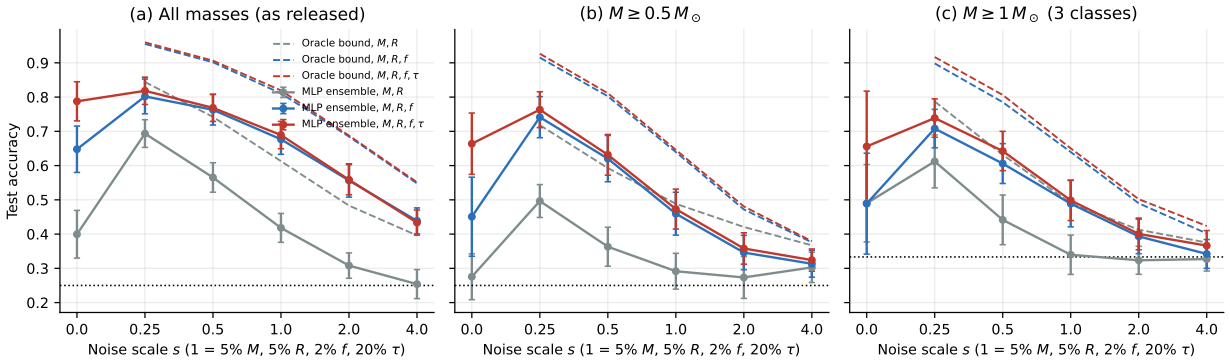


Figure 11: Single-sequence data: test accuracy versus noise scale s for the MLP ensemble (solid, 95% t-intervals over 10 repetitions) and the oracle Bayes bound (dashed). Dotted: chance.

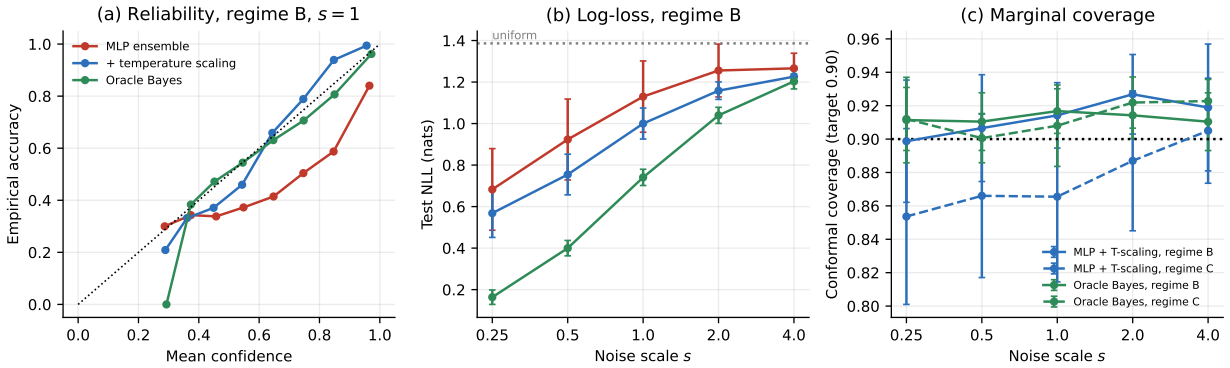


Figure 12: Single-sequence data: (a) reliability diagram (regime B, $s = 1$), (b) NLL, (c) conformal coverage versus noise scale.

B Experimental settings

Table 14: Settings.

Item	Setting
Population folds	5 folds by EoS (stratified), test / calibration / training = 1 / 1 / 3 folds, 3 random partitions
Population samples	training: 33 masses × 3 noise draws per EoS; calibration: 33 × 2; test: 33 × 3 (single draw when noise-free)
MLP	3 members, 2×64 ReLU, Adam, lr 10^{-3}, $\alpha = 10^{-4}$, batch 128, early stopping (max 120 iterations)
Other learners	logistic regression (L2); k-NN, $k = 25$; random forest, 200 trees, min leaf 5; histogram gradient boosting (150 iterations); RBF SVM, $C = 3$, 8000 samples
Kernel Bayes	reference: 22 masses per EoS; training EoS, or all non-test EoS
Robustness studies	EoS-level bootstrap: 2000 resamples within class; learning curves: 25, 50, 75, 100% of the training EoS; population B: 48 EoS per scenario, same folds protocol; out-of-family: train on all EoS of one population, test on all of the other, three noise seeds
Numerical tests	integration tolerances 10^{-8}–10^{-13}, surface cut-off 10^{-6}–10^{-4}, centre radius ×0.5–2.5, matching radius 1500–8000 km, crust pressure scale ±10%
Conformal	LAC score, $\alpha = 0.1$, correction with the number of calibration EoS
Single-sequence study	blocks of six models, 60/20/20, 10 repetitions; 20 noise draws per training model; random forest 150 trees; 15 repetitions for the block-length study, 100 for the random-split replication

Table 13: Single-sequence data: test accuracy (%, mean ± 95% t-interval over 10 repetitions) by regime, observable set and noise scale s. Chance is 25% (A, B) or 33% (C).

Regime	Observables	s	MLP ensemble	Logistic reg.	Random forest	Oracle bound
A	M, R	0.25	69.4±4.0	35.9±5.4	21.3±3.5	84.4±2.7
A	M, R	1	41.8±4.2	37.1±4.5	24.8±3.5	61.3±4.2
A	M, R	4	25.4±4.2	29.1±3.7	23.1±2.5	39.6±4.1
A	M, R, f	0.25	80.2±5.1	38.7±4.9	56.0±4.9	95.5±1.1
A	M, R, f	1	67.7±4.4	38.7±5.2	56.3±3.5	81.1±2.8
A	M, R, f	4	43.9±3.8	38.3±4.3	39.5±2.7	54.8±3.3
A	M, R, f, τ	0.25	81.8±4.0	44.8±4.8	58.9±3.7	96.0±1.0
A	M, R, f, τ	1	68.9±4.0	40.3±4.3	59.6±3.8	81.9±2.8
A	M, R, f, τ	4	43.4±3.7	37.8±4.2	39.4±2.9	55.1±3.1
B	M, R	0.25	49.7±4.8	40.6±6.0	30.9±4.7	71.8±3.9
B	M, R	1	29.2±5.2	35.9±5.7	28.1±3.5	48.8±2.4
B	M, R	4	30.3±4.4	32.1±4.3	30.1±1.6	36.7±2.5
B	M, R, f	0.25	74.1±6.0	59.5±5.2	40.8±5.8	91.5±1.9
B	M, R, f	1	46.0±6.3	45.5±4.3	38.8±3.8	63.9±2.5
B	M, R, f	4	31.3±3.9	35.1±4.3	30.6±2.6	37.6±2.2
B	M, R, f, τ	0.25	76.3±5.2	63.6±3.9	43.4±5.7	92.7±1.8
B	M, R, f, τ	1	47.3±5.8	46.4±4.4	39.9±4.3	64.6±2.7
B	M, R, f, τ	4	32.4±3.2	35.4±4.2	31.8±2.2	37.9±2.1
C	M, R	0.25	61.2±7.7	62.6±7.5	37.7±8.7	78.7±4.2
C	M, R	1	34.0±5.7	40.7±7.1	33.1±5.5	49.0±4.8
C	M, R	4	32.7±3.5	33.9±6.2	33.2±2.5	37.5±5.8
C	M, R, f	0.25	70.8±5.6	69.0±6.7	44.1±10.0	89.8±1.6
C	M, R, f	1	48.9±6.8	51.4±6.6	41.9±6.7	64.0±3.5
C	M, R, f	4	34.2±4.2	36.0±5.1	34.3±3.4	40.2±4.3
C	M, R, f, τ	0.25	73.9±5.6	72.5±6.4	50.0±9.6	91.7±1.4
C	M, R, f, τ	1	49.9±5.9	52.5±7.0	44.1±6.9	65.0±3.1
C	M, R, f, τ	4	36.6±4.4	36.6±5.3	35.7±3.1	42.4±4.7

C Changes relative to arXiv:2512.23758v3

Table 15: Statements of the earlier version that are withdrawn or changed.

Earlier statement	Status
Each class is one EoS; the classifier "separates composition"	Replaced by populations of EoS with whole-EoS hold-out; the accuracy ceiling is about 64% and nucleonic and dark-matter-admixed matter are not distinguishable.
≈ 190 models per class, 770 in total	Total is correct; classes contain 222, 220, 122 and 206 models.
Models span $\sim 1\,M_\odot$ to the maximum mass	The data span 7×10^{-8} to $2.16\,M_\odot$; 70% have $M < 1\,M_\odot$.
97.4% test accuracy	Reproducible (97.7% average over 100 splits) but inflated by neighbouring models; falls to 94–69% with six- to 24-model blocks.
Per-class accuracy ranges 82–97% "across dataset realizations"	Not reproduced; removed.
"Mild degradation near the high-mass end"	Not supported by the earlier figure; removed.
"Not merely overfitting noise"	The inputs were noise-free; replaced by explicit error scenarios.
Permutation importance shows f and τ dominate	Replaced by observable-set ablations: f carries the information, τ adds nothing.
Tabulated redshift used as an input	Inconsistent with (M, R) for two classes; removed.
Oscillation quantities from the earlier code	Population data generated with a new solver validated against six independent f-mode frequencies.

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